

# QUANTITATIVE CHEMISTRY – THE MOLE

## KEY VOCABULARY

**Mole:** the unit for amount of substance ( $6.02 \times 10^{23}$  particles).

**Avogadro constant:** the number of particles in one mole,  $6.02 \times 10^{23}$ .

**Molar mass:** the  $M_r$  in grams — the mass of one mole.

**Amount (mol):** calculated from  $\text{mass} \div \text{molar mass}$ .

## COMMON MISCONCEPTIONS

~~— A mole is a unit of mass.~~

✓ A mole is an amount (a count of particles).

~~— One mole always weighs the same.~~

✓ The mass of a mole depends on the molar mass.

~~— Avogadro's number changes per element.~~

✓ It is always  $6.02 \times 10^{23}$  particles per mole.

## MOLES FORMULA

$$\text{moles} = \frac{\text{mass (g)}}{\text{molar mass (g/mol)}}$$

## CONVERSION REMINDER

$$\text{number of particles} = \text{moles} \times 6.02 \times 10^{23}$$

### 1 MASS → MOLES

Calculate the moles in 36 g of water. ( $M_r = 18$ ) [2 marks]

Answer: \_\_\_\_\_ mol

### 2 MOLES → MASS

Calculate the mass of 0.5 mol of  $\text{CaCO}_3$ . ( $M_r = 100$ ) [2 marks]

Answer: \_\_\_\_\_ g

### 3 PARTICLES

Calculate the number of molecules in 2 mol of  $\text{CO}_2$ . [2 marks]

Answer: \_\_\_\_\_ molecules

### 4 DEFINE A MOLE

State what is meant by **one mole** of a substance. [1 mark]

\_\_\_\_\_

### 5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

To find moles, \_\_\_\_ the \_\_\_\_ by the molar mass.

mass

molar mass

multiply

divide

\_\_\_\_\_

### 6 COMPARE

Calculate the moles in 23 g of sodium. ( $A_r = 23$ ) [1 mark]

Answer: \_\_\_\_\_ mol